Systematic Underestimation in Defense Cost and Schedule Estimates

Evidence from GAO's 2024 and 2025 Weapon Systems Annual Assessments

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Executive Summary

The Department of Defense (DoD) plans and budgets its largest weapon programs around cost and schedule estimates made early in development, when uncertainty is highest. When those early estimates are too optimistic, the result is funding shortfalls, delayed capabilities, and difficult tradeoffs later. This analysis asks whether DoD's early estimates are consistently wrong in the same direction and, if so, how much adjustment a decision-maker should apply.

Using program-level data from GAO's 2024 and 2025 Weapon Systems Annual Assessments, this analysis compared the first full estimate with the current estimate for 31 major defense acquisition programs (MDAPs), in constant fiscal year 2025 dollars. Three findings stand out:

1. **Schedule estimates miss in only one direction.** Of 22 programs with schedule data, 20 have slipped, 2 are on schedule, and none is ahead of schedule. The median program's time to initial capability has grown 38%. Between the 2024 and 2025 reports, 12 of these programs slipped further and none improved.

2. **Schedule growth appears to precede cost growth.** Programs whose first estimate was made in 2016 or earlier show a median unit cost increase of 27% and schedule increase of 51%. Newer programs are close to their original cost (-1%) but have already slipped 29% on schedule.

3. **A reference class adjustment is large.** Based on these programs' history, a schedule estimate would need about 1.4 times the original plan to have a 50% chance of holding, and about 1.9 times for 80% confidence.

These results are consistent with decades of research on optimism bias and strategic misrepresentation in large projects. The main recommendations are to report schedule estimates as ranges with stated confidence levels, building on existing cost-risk requirements, and to check program estimates against how similar programs actually performed (reference class forecasting). This is a first-pass analysis using public data: the sample is small, most programs are not yet complete, and the data cannot separate estimating error from changes in scope.

Background

The cost of optimistic estimates

Each year, GAO assesses DoD's costliest weapon programs. Its 2025 assessment reviewed 106 programs and found that the combined cost estimates for 30 MDAPs increased by $49.3 billion in a single year. One program, the LGM-35A Sentinel intercontinental ballistic missile, accounted for more than $36 billion (73%) of that increase. Schedules also slipped: the average time to deliver initial capability grew by 18 months, to almost 12 years from program start.

Why estimates may run low

Research on large projects outside defense finds a similar pattern. In a study of 258 transport infrastructure projects, Flyvbjerg, Holm, and Buhl (2002) found that costs were underestimated in about 9 of 10 projects, with actual costs about 28% higher than estimated on average. Because the errors were overwhelmingly in one direction, the authors concluded they could not be explained by random error.

Two explanations are commonly offered:

- **Optimism bias (the planning fallacy).** Kahneman and Tversky (1979) described a tendency to base forecasts on a best-case plan for the specific project. Kahneman and Lovallo (1993) later called this the "inside view," as opposed to the "outside view" of how similar projects actually turned out. The people producing the estimate genuinely believe it.

- **Strategic misrepresentation.** Flyvbjerg and colleagues argue that estimates can also be kept low deliberately, because optimistic numbers help projects win approval and funding.

Both explanations predict the same observable pattern: errors that lean consistently toward underestimation.

Reference class forecasting

Reference class forecasting, developed by Flyvbjerg (2006) from Kahneman's work on the outside view, corrects for both explanations without needing to determine which one applies. Instead of relying only on a program's own estimate, the forecaster looks at the distribution of outcomes for a group of similar past programs and adjusts the new estimate to reach a desired level of confidence.

Problem Statement and Research Questions

DoD decision-makers rely on early cost and schedule estimates to make funding, portfolio, and affordability decisions. If those estimates are systematically optimistic, decisions based on them carry more risk than the numbers suggest, and problems surface only after significant resources are committed.

This analysis addresses three questions:

1. Do cost and schedule estimates for major defense programs consistently miss in the same direction?

2. Does the pattern differ between older and newer programs?

3. Based on historical outcomes, what adjustment would a reference class approach suggest for a new estimate?

Data and Methodology

Data sources

- **GAO-24-106831,** *Weapon Systems Annual Assessment* (June 2024), FY2024 dollars
- **GAO-25-107569,** *Weapon Systems Annual Assessment* (June 2025), FY2025 dollars
- **DoD Selected Acquisition Report (SAR) Summary Tables**, December 2019, used for context on the causes of cost growth.

Program selection

Data were taken from the "Program Performance" table in GAO's two-page MDAP assessments, which report the first full estimate, the prior year's estimate, and the current estimate for total acquisition cost, unit cost, quantities, and cycle time (months to initial capability). The following were excluded:

- Middle Tier of Acquisition (MTA) programs, which do not report a first full estimate
- MDAP increments and future major weapon acquisitions (one-page assessments without comparable data)
- Programs with no formal estimate yet, or whose first estimate is also their current estimate (no growth history)

The final dataset includes **31 programs** with a first estimate to compare against, of which **22** report schedule data in both estimates. Where the 2025 report no longer listed a program's first full estimate, it was taken from the 2024 report. The F-35 program, not assessed in 2025, is included using its 2024 figures.

Measures

For each program, growth was measured as a ratio:
Growth ratio = current estimate ÷ first full estimate

A ratio of 1.0 means no change, and 1.3 means 30% growth. This was calculated for total acquisition cost, unit cost (total cost ÷ quantity), and cycle time. Unit cost is emphasized because total cost is distorted by changes in the number of units purchased.

Inflation adjustment

All values are in constant FY2025 dollars. FY2024 values were converted by a factor of 1.023. This factor was verified against five programs whose first full estimates appear in both reports and differ by exactly 2.3%.

Reference class calculation

Schedule growth ratios were sorted to form an empirical cumulative distribution. The 50th and 80th percentiles give the adjustment a new schedule estimate would need to achieve 50% and 80% confidence, based on these programs' outcomes.

Statistical checks

A one-sided sign test was applied to schedule changes, excluding the two programs with no change. If slips and early completions were equally likely, the probability of all 20 changes being slips is 0.5^{20}, or about 1 in 1 million. To assess the uncertainty of the 80th percentile, the 22 schedule ratios were resampled with replacement 20,000 times (bootstrap), and the middle 95% of the resulting 80th percentile values was taken. Both methods assume programs are independent, which is only approximately true.

Tools

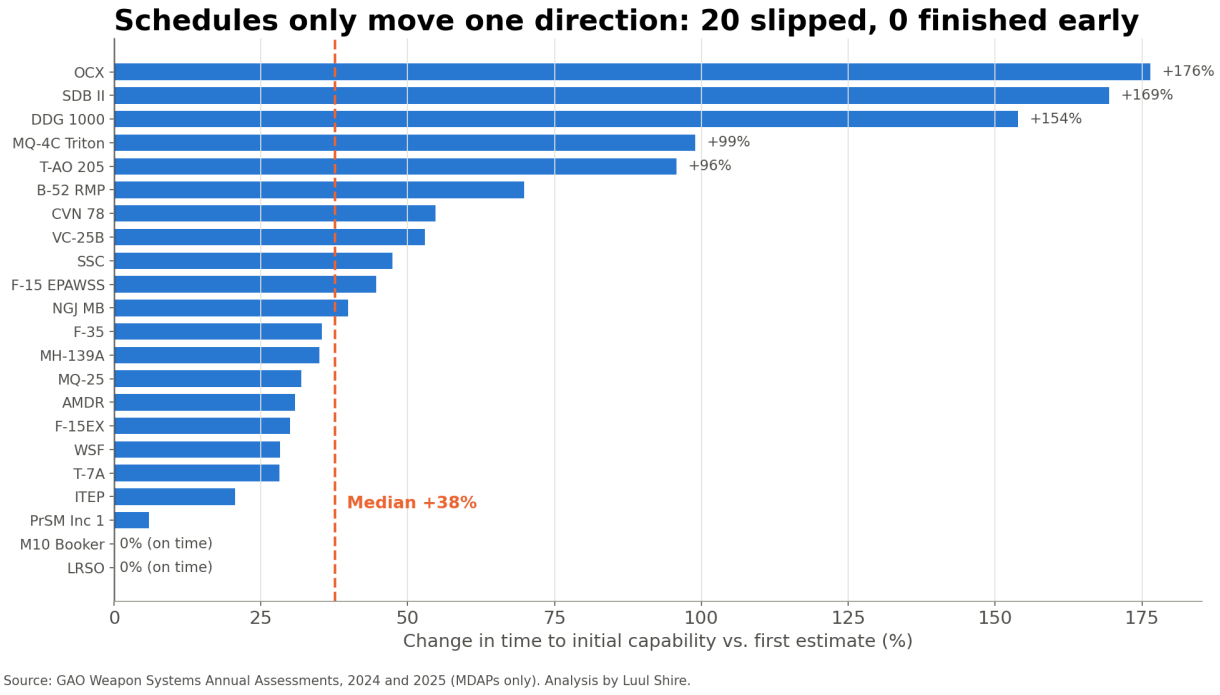
Data was compiled manually from the GAO reports and analyzed in Python using pandas, NumPy, and Matplotlib.

Findings

Finding 1: Schedule estimates only miss in one direction

Of the 22 programs with schedule data, **20 have slipped, 2 are on schedule, and none is ahead of schedule.** A sign test indicates this one-directional pattern is very unlikely to be due to chance ($p \approx 0.000001$). The median program's time to initial capability has grown **38%** beyond its first estimate. The largest slips include OCX (+176%), SDB II (+169%), and DDG 1000 (+154%).

The slips are also continuing. Comparing the same programs between the 2024 and 2025 reports, **12 programs slipped further, 8 held steady, and none recovered time.** Several programs that appeared on track in 2024 slipped in 2025, including B-52 RMP, whose cycle time grew from 73 to 107 months in one year.



Source: GAO Weapon Systems Annual Assessments, 2024 and 2025 (MDAPs only). Analysis by Luul Shire.

Figure 1. Change in time to initial capability vs. first estimate, by program.

Finding 2: Schedule growth appears to precede cost growth

Across the **30 programs with unit cost data** (MGUE Increment 1 does not report a unit quantity), cost results are mixed at first glance: 16 programs are above their original unit cost and 14 are below, with a median change of +2%. But the picture changes when programs are grouped by when their first estimate was made:

TABLE 1

First estimate made	Programs	Median schedule change	Median unit cost change
2016 or earlier	11	+51%	+27%
2017 or later	20	+29%	-1%

Newer programs have already slipped substantially on schedule but have not yet shown cost growth. Older programs show both. This suggests that schedule slips may be an early warning, with cost growth surfacing later, often in production. Because this compares older and newer programs at a single point in time rather than tracking the same programs over time, the pattern should be confirmed with multi-year data. It also suggests that the near-flat cost picture for newer programs may understate their eventual growth.

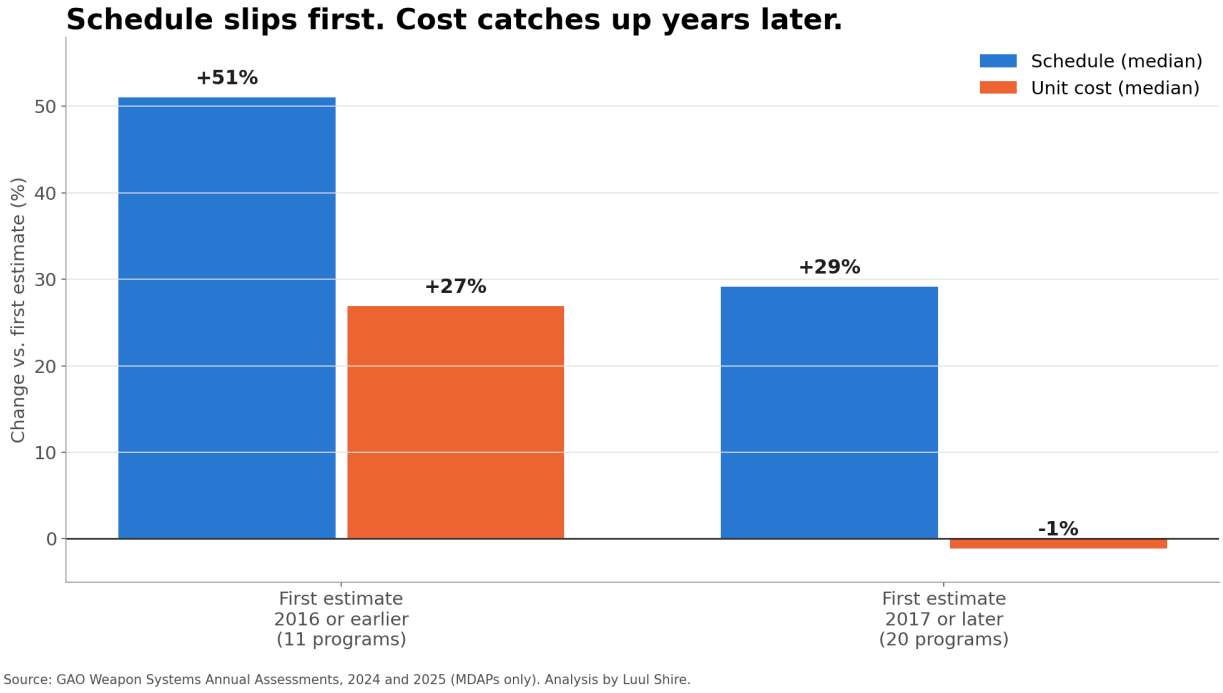


Figure 2. Median schedule and unit cost change, by age of first estimate.

Finding 3: A reference class adjustment is large

Using the distribution of schedule outcomes for these 22 programs:

- **50% confidence:** plan for about **1.38 times** the original schedule estimate
- **80% confidence:** plan for about **1.91 times** the original schedule estimate

For example, a program estimating 100 months to initial capability would, based on this history, need to plan for about **191 months** to have an 80% chance of meeting its date. Because the 1.91 figure rests on only 22 programs, it is uncertain: a bootstrap analysis gives a 95% range of about 1.5 to 2.7 times. Even the low end implies a substantial schedule buffer.

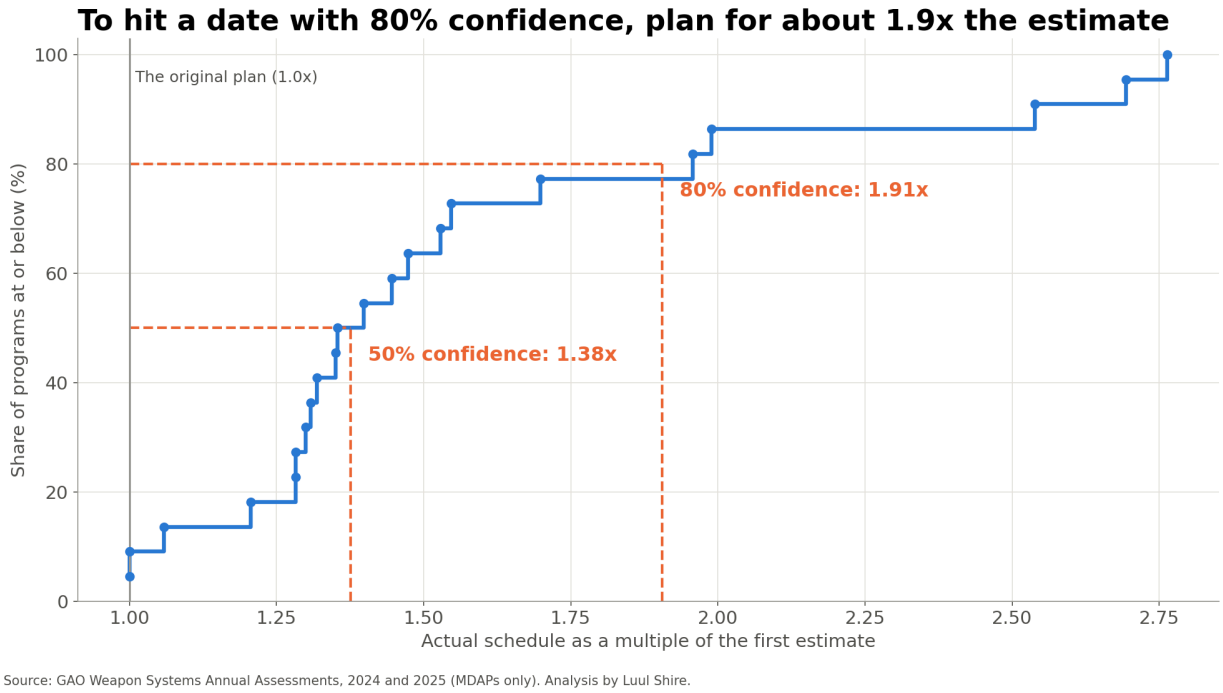


Figure 3. Share of programs finishing within a given multiple of their first schedule estimate.

Case example: Sentinel

Sentinel illustrates the gap between a program's own estimate and an independent one. Its first full estimate in September 2020 was $94.1 billion. After a critical Nunn-McCurdy breach in January 2024, the program's estimate rose to $129.0 billion, 37% higher. DoD's independent cost office (CAPE) estimated that the modified program could cost about $170.6 billion, roughly 81% more than the original estimate. The independent estimate exceeds the program's own estimate by $41.6 billion.

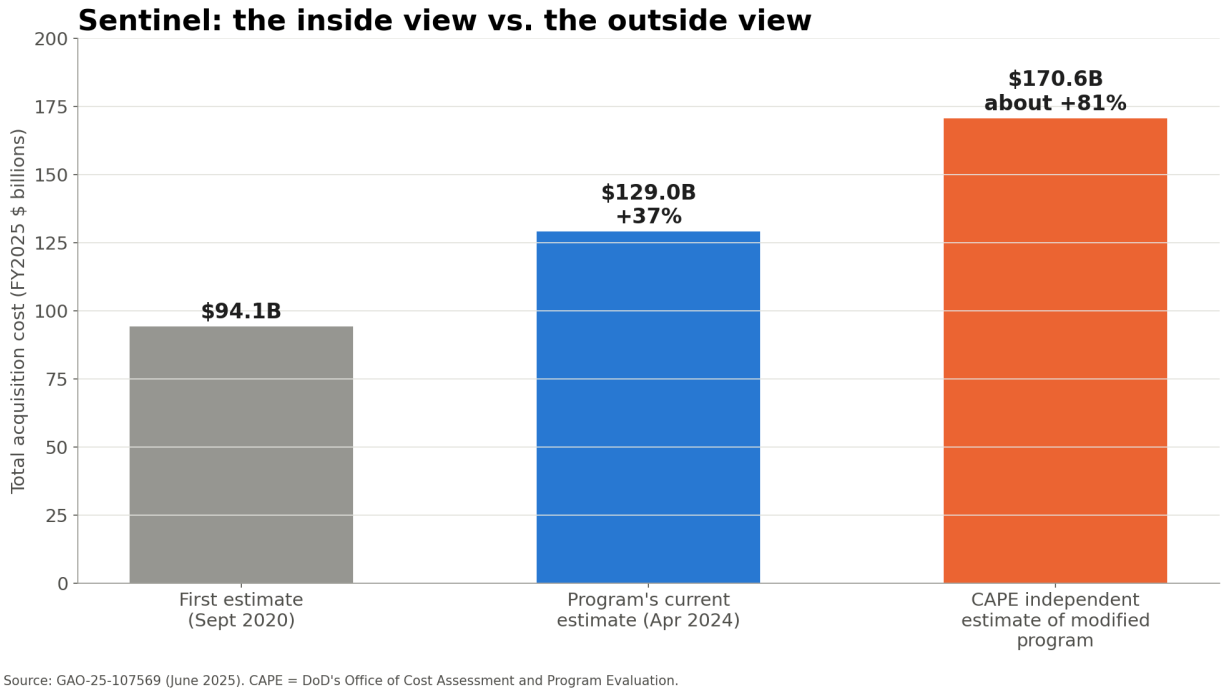


Figure 4. Sentinel: first estimate, current program estimate, and CAPE independent estimate (FY2025 $ billions).

Discussion

Reported cost decreases can be misleading

Several programs reported cost decreases that, according to GAO, did not reflect real savings:

- **Inflation or index updates:** SSBN 826 (2024), T-AO 205, GPS IIIF, and LRSO
- **Accounting changes:** T-7A removed its inflation contingency; PrSM shifted software costs to another budget
- **Quantity cuts:** DDG 1000 (32 ships reduced to 3, with unit cost about 7 times the original), MQ-4C Triton, and MH-139A
- **Excluded costs:** FFG 62's estimate does not include pending contractor claims, while its lead ship is about three years late

In addition, several programs used fixed-price contracts (KC-46A, VC-25B, and T-7A), under which much of any overrun falls on the contractor rather than the government. On these programs, the government's estimate may understate the true cost growth.

Together, these suggest that official cost estimates may understate growth, which makes the schedule data a more reliable signal.

What DoD attributes cost growth to

DoD's SAR data offer an important counterpoint. Across all programs reported in December 2019, DoD attributed most cost growth since baseline to **quantity changes (+$123.4 billion)** and **engineering changes (+$78.8 billion)**. The "estimating" category, which captures corrections to the estimate itself, was **net negative (-$38.2 billion).**

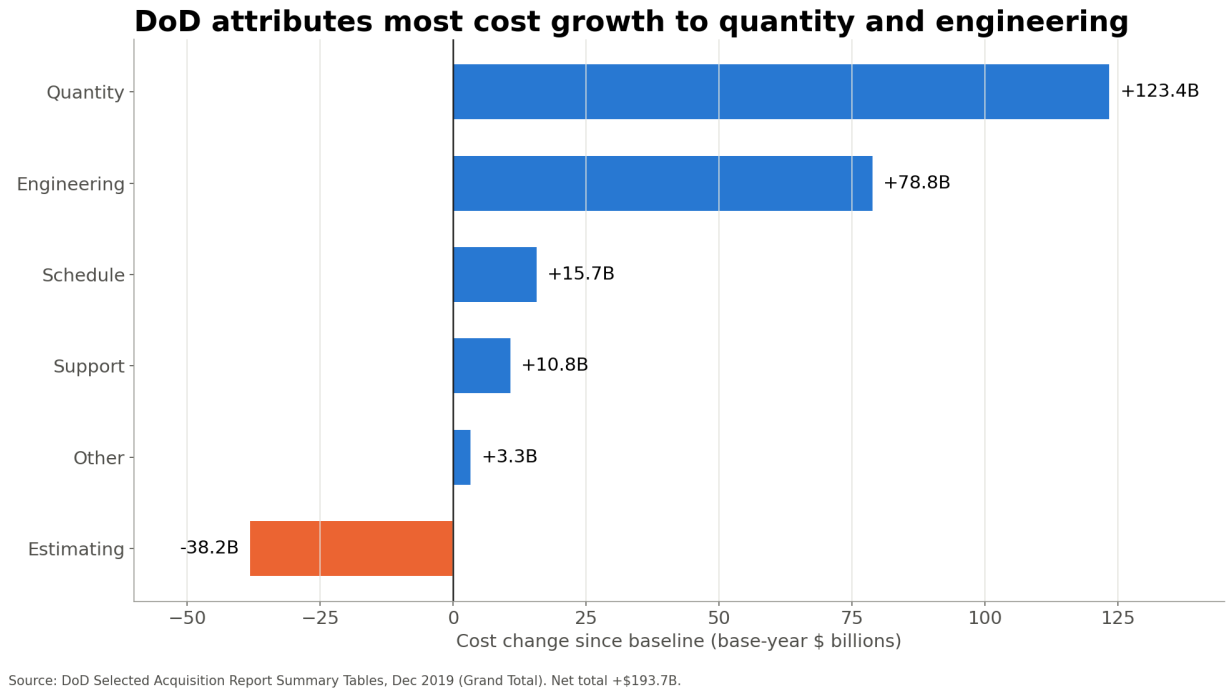


Figure 5. Cost change since baseline by cause, DoD SAR Summary Tables, December 2019 (base-year $ billions).

This means cost growth cannot be attributed simply to "bad estimates." However, engineering changes often reflect optimistic technical assumptions made early, such as immature technologies that later require redesign. GAO's assessments repeatedly cite immature technology at development start (for example, Sentinel and FLRAA). In one program, officials attributed technical problems partly to "the vendors' underestimation of the complexity of the development effort" (MGUE Increment 2). Schedule, which is less affected by changes in quantity, shows the clearest one-directional pattern.

Early optimism appears even in new programs

Two programs with recent baselines had already slipped within months: T-AGOS 25's schedule grew from 64 to 97 months within seven months of its first estimate, and Next Gen OPIR GEO's grew from 86 to 89 months shortly after its baseline was approved. This suggests optimism is present at the time estimates are set, not only as programs age.

Recommendations

1. **Report schedule estimates as ranges.** DoD guidance already requires a discussion of risk in cost estimates (10 U.S.C. § 3225, originating in WSARA 2009), and GAO's Cost Estimating and Schedule Assessment Guides call for risk analysis. Extending this practice to schedule, with 50% and 80% confidence dates rather than a single date, would make schedule risk explicit.

2. **Anchor estimates to past program outcomes.** Alongside CAPE's independent cost estimate, a reference class check would compare each new estimate against the historical growth of similar programs.

3. **Treat schedule slips as an early warning for cost growth.** Because schedule growth appears to precede cost growth, programs with early schedule slips could be flagged for closer cost review before growth appears in the budget.

Limitations and Future Work

- **Small sample:** 31 programs (22 with schedule data) limit how precise these estimates can be, and the results should not be applied to any single program without judgment.

- **Current estimates, not final costs:** most programs are still in progress, so final growth is likely higher than shown.

- **Two-year snapshot:** GAO reports only the first, prior-year, and current estimates. A longer history would allow stronger analysis.

- **Mixed causes:** the data cannot separate estimating error from changes in scope, quantity, or requirements. Unit cost reduces, but does not remove, the effect of quantity changes.

- **Different program types:** ships, aircraft, missiles, and software systems may have different growth patterns. A larger dataset would allow separate reference classes by type.

- **Point-in-time comparison:** the older-versus-newer comparison does not follow the same programs over time, and the statistical tests assume programs are independent.

Future work could use DoD's full SAR history to build reference classes by program type, separate growth by cause, and test whether adjusted estimates would have predicted outcomes better than original estimates for completed programs.

References

- Flyvbjerg, B. (2006). From Nobel Prize to project management: Getting risks right. *Project Management Journal,* 37(3), 5-15.

- Flyvbjerg, B., Holm, M.S., & Buhl, S. (2002). Underestimating costs in public works projects: Error or lie? *Journal of the American Planning Association,* 68(3), 279-295.

- Kahneman, D., & Lovallo, D. (1993). Timid choices and bold forecasts: A cognitive perspective on risk taking. *Management Science,* 39(1), 17-31.

- Kahneman, D., & Tversky, A. (1979). Intuitive prediction: Biases and corrective procedures. *TIMS Studies in Management Science*, 12, 313-327.

- U.S. Department of Defense. (2020). *Selected Acquisition Report (SAR) Summary Tables*, December 2019.

- U.S. Government Accountability Office. (2015). *Schedule Assessment Guide: Best Practices for Project Schedules* (GAO-16-89G).

- U.S. Government Accountability Office. (2020). *Cost Estimating and Assessment Guide: Best Practices for Developing and Managing Program Costs* (GAO-20-195G).

- U.S. Government Accountability Office. (2024). *Weapon Systems Annual Assessment* (GAO-24-106831).

- U.S. Government Accountability Office. (2025). *Weapon Systems Annual Assessment* (GAO-25-107569).

- Weapon Systems Acquisition Reform Act of 2009, Pub. L. No. 111-23, 123 Stat. 1704; 10 U.S.C. § 3225 (formerly § 2334), as amended by Pub. L. No. 114-328 (2016).